

MODEL comparision matrix Validation

Classification	Accuracy	Precision	Recall	F1Score
Logistic Regression	0.734	0.126	0.757	0.215
Decision Tree Classifier	0.914	0.032	0.027	0.029
Random Forest Classifier	0.949	0	0	0
Gradient Boosting Classifier (e.g., XGBoost or LightGBM)	0.948	0	0	0
K-Nearest Neighbors Classifier	0.948	0.2	0.027	0.048
Support Vector Classifier (SVC)	0.672	0.118	0.892	0.208
Voting Vs Avearge (best 3 out of 6)	0.785	0.081	0.55	0.141
Ensemble	0.8	0.1	0.6	0.17

MODEL comparision matrix Test

Classification	Accuracy	Precision	Recall	F1Score
Logistic Regression	0.704	0.117	0.763	0.204
Decision Tree Classifier	0.914	0.111	0.105	0.108
Random Forest Classifier	0.949	0	0	0
Gradient Boosting Classifier (e.g., XGBoost or LightGBM)	0.949	0.333	0.026	0.049
K-Nearest Neighbors Classifier	0.95	0	0	0
Support Vector Classifier (SVC)	0.644	0.104	0.816	0.185
Voting Vs Avearge (best 3 out of 6)	0.766	0.185	0.535	0.146
Ensemble	0.78	0.2	0.58	0.17

ROC/AUC
0.838
0.493
0.802
0.832
0.63
0.806
0.825
0.84

ROC/AUC
0.827
0.531
0.767
0.81
0.602
0.805
0.814
0.83